

Path 2: Cyber Security 101 > Module 5: Networking > Room 2: Networking Essentials

<https://tryhackme.com/room/networkingessentials>

Networking Essentials:

Explore networking protocols from automatic configuration to routing packets to the destination.

>> Task 1: Introduction

Have you ever wondered how your computer can dynamically configure its network settings when you turn it on or connect it to a new network? Have you ever wanted to know how many devices and countries your packets passed through before reaching their destination? Are you curious how all your home devices can access the Internet even though your ISP gives you a single IP address?

♦ Learning Prerequisites:

To benefit from this room, we recommend that you know the following:

- ISO OSI model and layers
- TCP/IP model and layers
- Ethernet, IP, and TCP protocols

In other words, starting this room after "Networking Concepts" is the recommended approach.

♦ Learning Objectives:

The objective of this room is to teach you about various standard protocols and technologies that glue things together:

- Dynamic Host Configuration Protocol (DHCP)
- Address Resolution Protocol (ARP)
- Network Address Translation (NAT)
- Internet Control Message Protocol (ICMP)
 - Ping
 - Traceroute

>> Task 2: DHCP: Give Me My Network Settings

DHCP: The Dynamic Host Configuration Protocol (DHCP) is a network management protocol used on Internet Protocol (IP) networks for automatically assigning IP addresses and other communication parameters to devices connected to the network using a client–server architecture.

You went to your favourite coffee shop, grabbed your favourite hot drink, and opened your laptop. Your laptop connected to the shop's WiFi and automatically configured the network, so you could now work on a new TryHackMe room. You didn't type a single IP address, yet your device is all set up. Let's see how this happened.

Whenever we want to access a network, at the very least, we need to configure the following:

- IP address along with subnet mask
- Router (or gateway)
- DNS server

Whenever we connect our device to a new network, the above configurations must be set according to the new network. Manually configuring these settings is a good option, especially for servers. Servers are not expected to switch networks; you don't carry your domain controller and connect it to the coffee shop WiFi. Moreover, other devices need to connect to the servers and expect to find them at specific IP addresses.

Having an automated way to configure connected devices has many advantages. First, it would save us from manually configuring the network; this is extremely important, especially for mobile devices. Secondly, it saves us from address conflicts, i.e., when two devices are configured with the same IP address. An IP address conflict would prevent the involved hosts from using the network resources; this applies to local resources and the Internet. The solution lies in using Dynamic Host Configuration Protocol (DHCP). DHCP is an application-level protocol that relies on UDP; the server listens on UDP port 67, and the client sends from UDP port 68. Your smartphone and laptop are configured to use DHCP by default.

D: Discover

O: Offer

R: Request

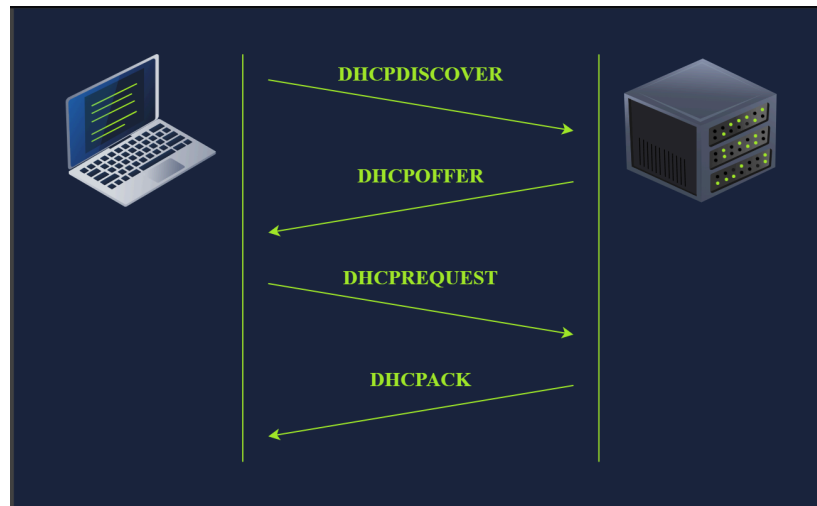
A: Acknowledge

DHCP follows four steps: Discover, Offer, Request, and Acknowledge (DORA).

:

- **DHCP Discover:** The client broadcasts a DHCPDISCOVER message seeking the local DHCP server if one exists.

- **DHCP Offer:** The server responds with a DHCPOFFER message with an IP address available for the client to accept.
- **DHCP Request:** The client responds with a DHCPREQUEST message to indicate that it has accepted the offered IP.
- **DHCP Acknowledge:** The server responds with a DHCPACK message to confirm that the offered IP address is now assigned to this client.



The following packet capture shows the four steps explained above. In this example, the client gets the address **192.168.66.133**.

user@TryHackMe\$ **tshark -r DHCP-G5000.pcap -n**

```

1  0.000000  0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Discover -
Transaction ID 0xfb92d53f
2  0.013904 192.168.66.1 → 192.168.66.133 DHCP 376 DHCP Offer  -
Transaction ID 0xfb92d53f
3  4.115318  0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request -
Transaction ID 0xfb92d53f
4  4.228117 192.168.66.1 → 192.168.66.133 DHCP 376 DHCP ACK   -
Transaction ID 0xfb92d53f
  
```

tshark -r DHCP-G5000.pcap -n is a Wireshark/TShark command used to read and display packets from a saved capture file.

-r = read

-n = Disable name resolution, Shows IP addresses and port numbers directly, Doesn't convert "IP to Domain Name" & "Port to Service Name".

In the DHCP packet exchange, we can notice the following:

- The client starts without any IP network configuration. It only has a MAC address. In the first and third packets, DHCP Discover and DHCP Request, the client searching for a DHCP server still has no IP network configuration and has not yet used the DHCP server's offered IP address. Therefore, it sends packets from the IP address **0.0.0.0** to the broadcast IP address **255.255.255.255**.
- As for the link layer, in the first and third packets, the client sends to the broadcast MAC address, **ff:ff:ff:ff:ff:ff** (not shown in the output above). The DHCP server offers an available IP address along with the network configuration in the DHCP offer. It uses the client's destination MAC address. (It used the proposed IP address in this example system.)

At the end of the DHCP process, our device would have received all the configuration needed to access the network or even the Internet. In particular, we expect that the DHCP server has provided us with the following:

- The leased IP address to access network resources
- The gateway to route our packets outside the local network
- A DNS server to resolve domain names (more on this later)

Q1) How many steps does DHCP use to provide network configuration?

Answer: 4

Q2) What is the destination IP address that a client uses when it sends a DHCP Discover packet?

Answer: 255.255.255.255

Q3) What is the source IP address a client uses when trying to get IP network configuration over DHCP?

Answer: 0.0.0.0

>> Task 3: ARP: Bridging Layer 3 Addressing to Layer 2 Addressing

ARP: Address Resolution Protocol (ARP) is responsible for finding the MAC (hardware) address related to a specific IP address. It works by broadcasting an ARP query, "Who has this IP address? Tell me." And the response is of the form, "The IP address is at this MAC address."

We have stated in the **Networking Concepts** room that as two hosts communicate over a network, an IP packet is encapsulated within a data link frame as it travels over layer 2. Remember that the two common data link layers we use are Ethernet (IEEE 802.3) and WiFi (IEEE 802.11). Whenever one host needs to communicate with another host on the same Ethernet or WiFi, it must send the IP packet within a data link layer frame. Although it knows the IP address of the target host, it needs to look up the target's MAC address so the proper data link header can be created.

As you would remember, a MAC address is a 48-bit number typically represented in hexadecimal notation; for example, **7C:DF:A1:D3:8C:5C** and **44:DF:65:D8:FE:6C** are two MAC addresses on my network.

However, the devices on the same Ethernet network do not need to know each other's MAC addresses all the time; they only need to know each other's MAC addresses while communicating. Everything revolves around IP addresses. Consider this scenario: You connect your device to a network, and if the network has a DHCP server, your device is automatically configured to use a specific gateway (router) and DNS server. Consequently, your device knows the IP address of the DNS server to resolve any domain name; moreover, it knows the IP address of the router when it needs to send packets over the Internet. In all this scenario, no MAC addresses are revealed. However, two devices on the same Ethernet cannot communicate without knowing each other's MAC addresses.

As a reminder, in the screenshot below, we see an IP packet within an Ethernet frame. The Ethernet frame header contains:

Destination MAC address

Source MAC address

Type (IPv4 in this case)

<https://tryhackme-images.s3.amazonaws.com/user-uploads/5f04259cf9bf5b57aed2c476/room-content/5f04259cf9bf5b57aed2c476-1719849170421.png>

♦ Address Resolution Protocol (ARP) makes it possible to find the MAC address of another device on the Ethernet. In the example below, a host with the IP address **192.168.66.89** wants to communicate with another system with the IP address **192.168.66.1**. It sends an ARP Request asking the host with the IP address **192.168.66.1** to respond. The ARP Request is sent from the MAC address of the requester to the broadcast MAC address, **ff:ff:ff:ff:ff:ff** as shown in the first packet. The ARP Reply arrived shortly afterwards, and the host with the IP address **192.168.66.1** responded with its MAC address. From this point, the two hosts can exchange data link layer frames.

```
user@TryHackMe$ tshark -r arp.pcapng -Nn
```

```
1 0.0000000000 cc:5e:f8:02:21:a7 → ff:ff:ff:ff:ff:ff ARP 42 Who has 192.168.66.1?
Tell 192.168.66.89
```

```
2 0.003566632 44:df:65:d8:fe:6c → cc:5e:f8:02:21:a7 ARP 42 192.168.66.1 is at
44:df:65:d8:fe:6c
```

```
tshark -r arp.pcapng -Nn
```

It reads the file arp.pcapng and displays packets in the terminal with numeric addresses only, without converting names.

-N:

- Controls other name resolution (like MAC → vendor name)
- When combined as **-Nn**, it keeps output mostly numeric/raw
- 👉 Result = No DNS / hostname / service name lookup

♦ If we use **tcpdump**, the packets will be displayed differently. It uses the terms ARP "Request" and ARP "Reply". For your information, the output is shown in the terminal below.

```
user@TryHackMe$ tcpdump -r arp.pcapng -n -v
```

```
17:23:44.506615 ARP, Ethernet (len 6), IPv4 (len 4), Request who-has 192.168.66.1
tell 192.168.66.89, length 28
```

```
17:23:44.510182 ARP, Ethernet (len 6), IPv4 (len 4), Reply 192.168.66.1 is-at
44:df:65:d8:fe:6c, length 28
```

```
"tcpdump -r arp.pcapng -n -v"
```

It reads the packet capture file arp.pcapng and prints packet details in the terminal with numeric addresses and extra verbose information.

-v: Verbose mode, Shows more packet details, Extra protocol fields and flags

An ARP Request or ARP Reply is not encapsulated within a UDP or even IP packet; it is encapsulated directly within an Ethernet frame. The following ARP Reply shows this.

<https://tryhackme-images.s3.amazonaws.com/user-uploads/5f04259cf9bf5b57aed2c476/room-content/5f04259cf9bf5b57aed2c476-1719849185596.png>

ARP is considered layer 2 because it deals with MAC addresses. Others would argue that it is part of layer 3 because it supports IP operations. What is essential to know is that ARP allows the translation from layer 3 addressing to layer 2 addressing.

Q1) What is the destination MAC address used in an ARP Request?

Answer: ff:ff:ff:ff:ff:ff

Q2) In the example above, what is the MAC address of 192.168.66.1?

Answer: 44:df:65:d8:fe:6c

>> Task 4: ICMP: Troubleshooting Networks

Internet Control Message Protocol (ICMP) is mainly used for network diagnostics and error reporting. Two popular commands rely on ICMP, and they are instrumental in network troubleshooting and network security. The commands are:

- **ping**: This command uses ICMP to test connectivity to a target system and measures the round-trip time (RTT). In other words, it can be used to learn that the target is alive and that its reply can reach our system.
- **traceroute**: This command is called **traceroute** on Linux and UNIX-like systems and **tracert** on MS Windows systems. It uses ICMP to discover the route from your host to the target.

♦ Ping:

You may have never played ping-pong (table tennis) before; however, thanks to ICMP, you can now play it with the computer! The **ping** command sends an ICMP Echo Request (ICMP Type 8). The screenshot below shows the ICMP message within an IP packet.

The screenshot shows a Wireshark packet capture window titled "Wireshark · Packet 1 · ICMP-ping.pcapng". The packet list on the left shows three packets: Frame 1 (98 bytes on wire), Ethernet II, and Internet Protocol Version 4. The selected packet (Frame 1) is expanded, showing the Internet Control Message Protocol details. The details pane shows the following information:

- Type: 8 (Echo (ping) request)
- Code: 0
- Checksum: 0x7288 [correct] [Checksum Status: Good]
- Identifier (BE): 3 (0x0003)
- Identifier (LE): 768 (0x0300)
- Sequence Number (BE): 1 (0x0001)
- Sequence Number (LE): 256 (0x0100)
- [Response frame: 2]
- Timestamp from icmp data: Jun 25, 2024 18:18:49.023154000 EEST
- [Timestamp from icmp data (relative): 0.000036287 seconds]
- Data (40 bytes)

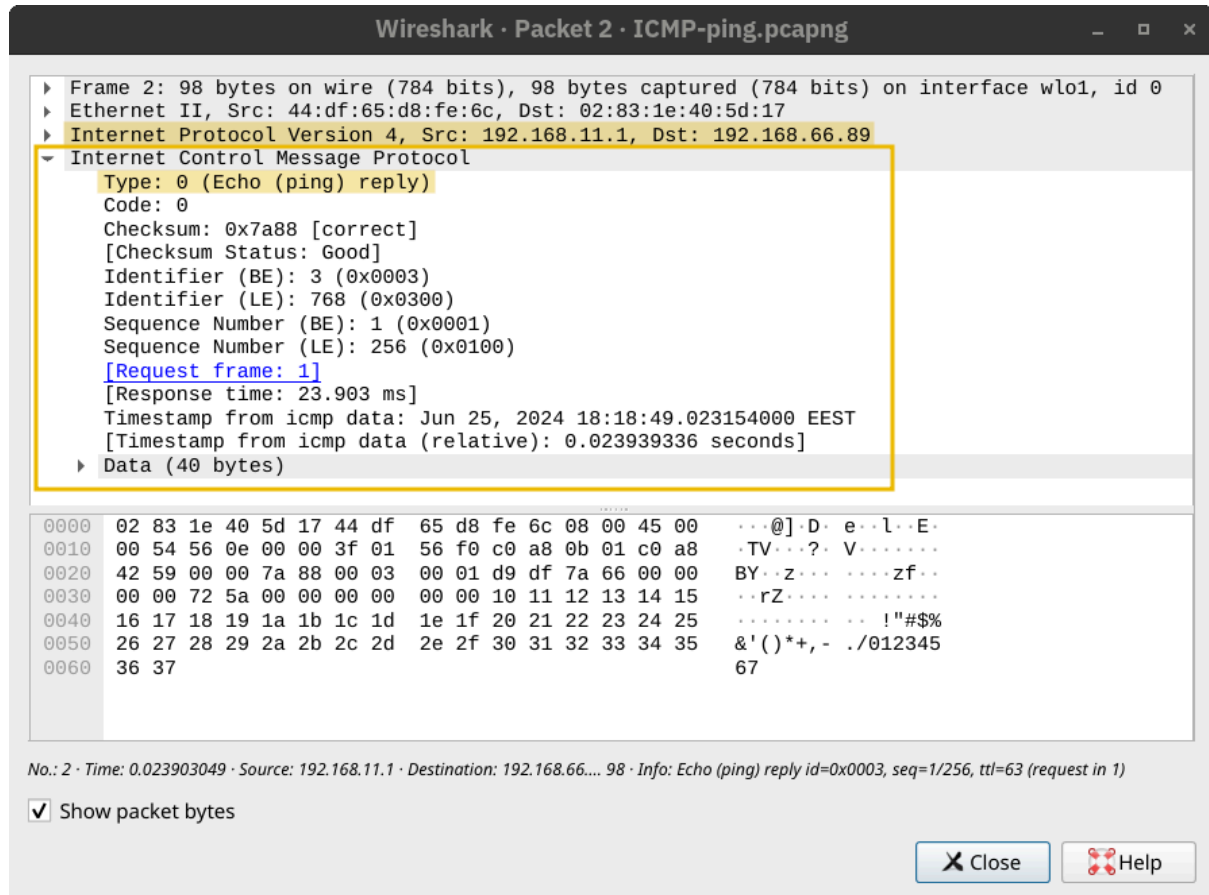
The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII. The ASCII column shows the string "D·e·l·@]...E·" followed by other characters.

No.: 1 · Time: 0.000000000 · Source: 192.168.66.89 · Destination: 192.168.11.1...: 98 · Info: Echo (ping) request id=0x0003, seq=1/256, ttl=64 (reply in 2)

☒ Show packet bytes

Close Help

The computer on the receiving end responds with an ICMP Echo Reply (ICMP Type **0**).



Many things might prevent us from getting a reply. In addition to the possibility of the target system being offline or shut down, a firewall along the path might block the necessary packets for **ping** to work. In the example below, we used **-c 4** to tell the **ping** command to stop after sending four packets.

Command: **ping 192.168.11.1 -c 4**

```
user@TryHackMe$ ping 192.168.11.1 -c 4
PING 192.168.11.1 (192.168.11.1) 56(84) bytes of data.
64 bytes from 192.168.11.1: icmp_seq=1 ttl=63 time=11.2 ms
64 bytes from 192.168.11.1: icmp_seq=2 ttl=63 time=3.81 ms
64 bytes from 192.168.11.1: icmp_seq=3 ttl=63 time=3.99 ms
64 bytes from 192.168.11.1: icmp_seq=4 ttl=63 time=23.4 ms

--- 192.168.11.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 3.805/10.596/23.366/7.956 ms
```

The output shows no packet loss; moreover, it calculates the minimum, average, maximum, and standard deviation (mdev) of the round-trip time (RTT).

- ♦ Traceroute:

How can we make every router between our system and a target system reveal itself?

The Internet protocol has a field called Time-to-Live (TTL) that indicates the maximum number of routers a packet can travel through before it is dropped. The router decrements the packet's TTL by one before it sends it across. When the TTL reaches zero, the router drops the packet and sends an ICMP Time Exceeded message (ICMP Type "11"). (In this context, "time" is measured in the number of routers, not seconds.)

The result of running **traceroute** to discover the routers between our system and **example.com**. Some routers don't respond; in other words, they drop the packet without sending any ICMP messages. Routers that belong to our ISP might respond, revealing their private IP address. Moreover, some routers respond and show their public IP address, and this would let us look up their domain name and discover their geographic location. Finally, there is always a possibility that an ICMP Time Exceeded message gets blocked and never reaches us.

Command: **traceroute example.com**

The traversed route might change as we rerun the command.

Q1) Using the example images above, how many bytes were sent in the echo (ping) request?

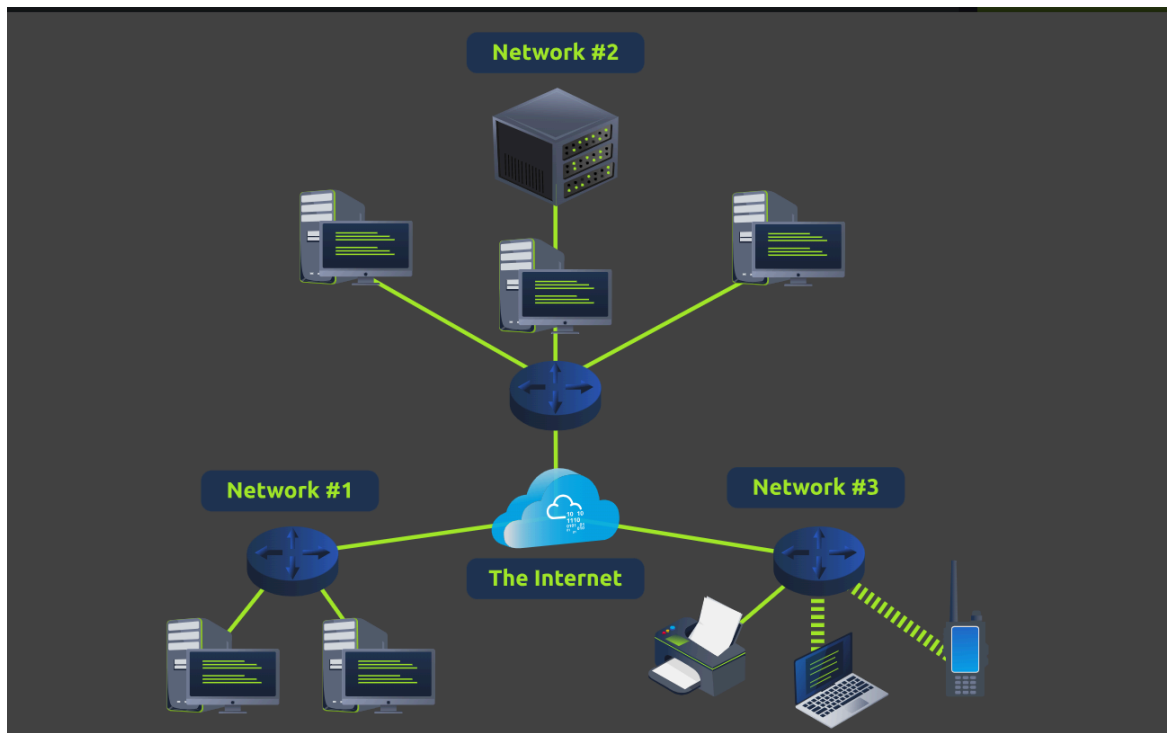
Answer: 40

Q2) Which IP header field does the traceroute command require to become zero?

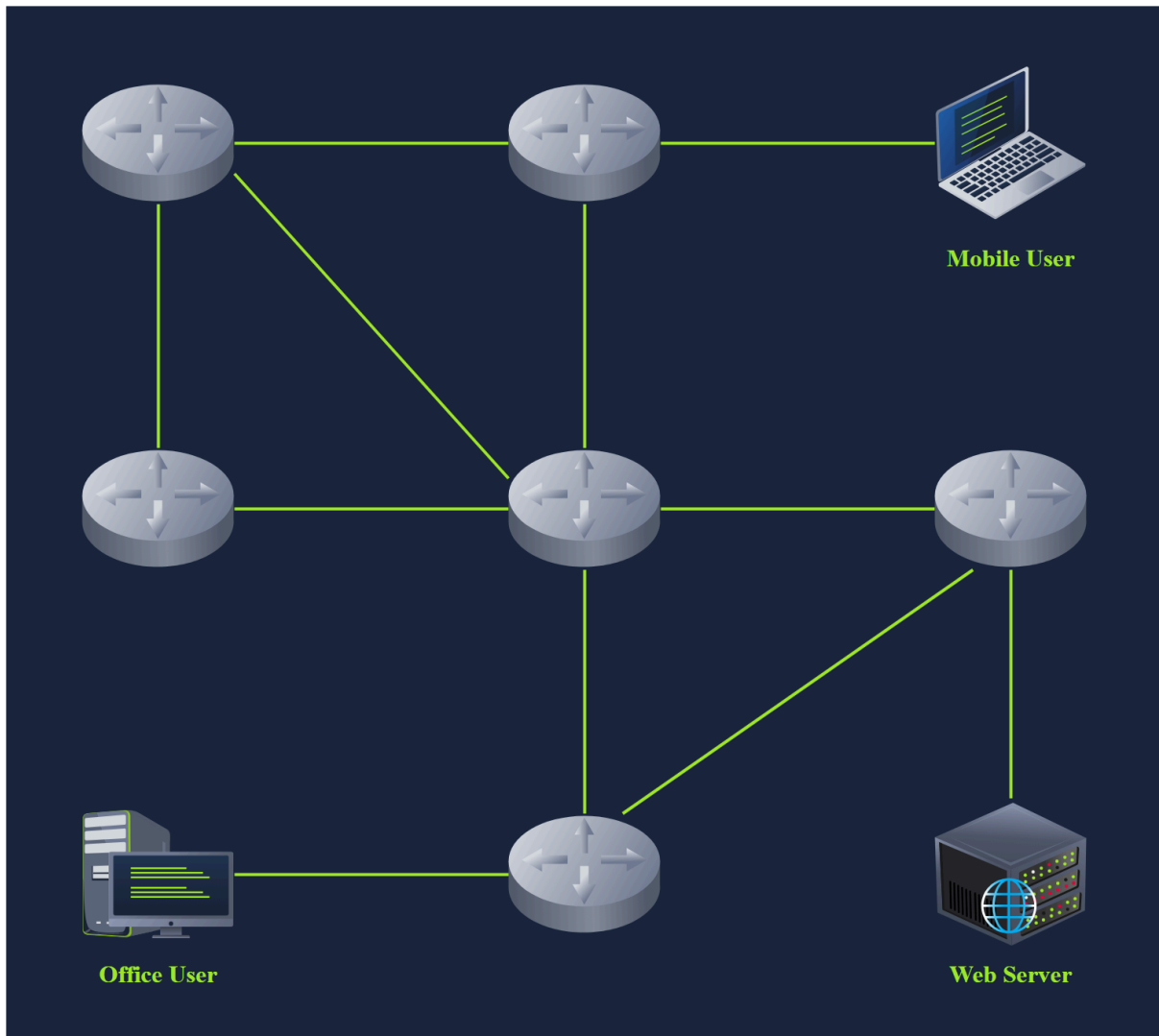
Answer: TTL

>> Task 5: Routing

Consider the network diagram shown below. It only has three networks; however, how can the Internet figure out how to deliver a packet from Network 1 to Network 2 or Network 3? Although this is an overly simplified diagram, we need some algorithm to figure out how to connect Network 1 to Network 2 and Network 3 and vice versa.



Let's consider a more detailed diagram. The Internet would be millions of routers and billions of devices. The network below is a tiny subset of the Internet. The mobile user can reach the web server; however, for this to happen, each router across the path needs to send the packets via the appropriate link. Obviously, there is more than one path, i.e., route, connecting the mobile user and the web server. We need a routing algorithm for the router to figure out which link to use.



The routing algorithms are beyond the scope of this room; however, we will briefly describe a few routing protocols so that you become familiar with their names:

- **OSPF (Open Shortest Path First):** OSPF is a routing protocol that allows routers to share information about the network topology and calculate the most efficient paths for data transmission. It does this by having routers exchange updates about the state of their connected links and networks. This way, each router has a complete map of the network and can determine the best routes to reach any destination.

- **EIGRP (Enhanced Interior Gateway Routing Protocol):** EIGRP is a Cisco proprietary routing protocol that combines aspects of different routing algorithms. It allows routers to share information about the networks they can reach and the cost (like bandwidth or delay) associated with those routes. Routers then use this information to choose the most efficient paths for data transmission.

- **BGP (Border Gateway Protocol):** BGP is the primary routing protocol used on the Internet. It allows different networks (like those of Internet Service Providers) to exchange routing information and establish paths for data to travel between these networks. BGP helps ensure data can be routed efficiently across the Internet, even when traversing multiple networks.

- **RIP (Routing Information Protocol):** RIP is a simple routing protocol often used in small networks. Routers running RIP share information about the networks they can reach and the number of hops (routers) required to get there. As a result, each router builds a routing table based on this information, choosing the routes with the fewest hops to reach each destination.

Q1) Which routing protocol discussed in this task is a Cisco proprietary protocol?

Answer: EIGRP

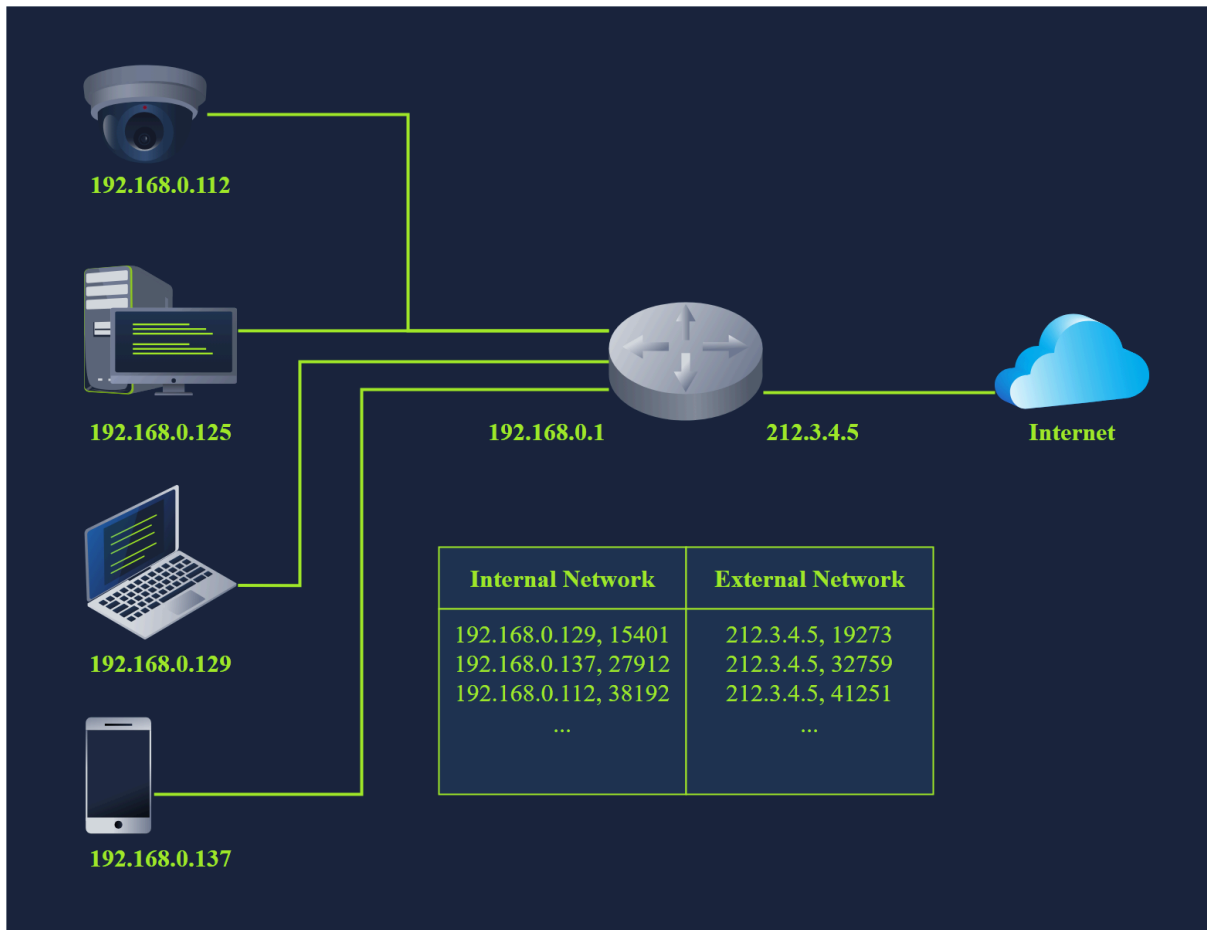
>> Task 6: NAT

As discussed in the **Networking Concepts** room, we calculated that IPv4 can support a maximum of four billion devices. With the increase in the number of devices connected to the Internet, from computers and smartphones to security cameras and washing machines, it was clear that the IPv4 address space would be depleted quickly. One solution to address depletion is Network Address Translation (NAT).

The idea behind NAT lies in using **one public IP address** to provide Internet access to **many private IP addresses**. In other words, if you are connecting a company with twenty computers, you can provide Internet access to all twenty computers by using a single public IP address instead of twenty public IP addresses. (Note: Technically speaking, the number of IP addresses is always expressed as a power of two. To be technically accurate, with NAT, you reserve two public IP addresses instead of thirty-two. Consequently, you would have saved thirty public IP addresses.)

Unlike routing, which is the natural way to route packets to the destination host, routers that support NAT must find a way to track ongoing connections. Consequently, NAT-supporting routers maintain a table translating network addresses between internal and external networks. Generally, the internal network would use a private IP address range, while the external network would use a public IP address.

In the diagram below, multiple devices access the Internet via a router that supports NAT. The router maintains a table that maps the internal IP address and port number with its external IP address and port number. For instance, the laptop might establish a connection with some web server. From the laptop perspective, the connection is initiated from its IP address **192.168.0.129** from TCP source port number **15401**; however, the web server will see this same connection as being established from **212.3.4.5** and TCP port number **19273**, as shown in the translation table. The router does this address translation seamlessly.



Q1) In the network diagram above, what is the public IP that the phone will appear to use when accessing the Internet?

Answer: 212.3.4.5

Q2) Assuming that the router has infinite processing power, approximately speaking, how many **thousand** simultaneous TCP connections can it maintain?

Answer: 65

>> Task 7: Closing Notes

Q1) I want to obtain an IP address to use and communicate with the other hosts on the network. What should I use?

Answer: DHCP

Q2) We need to give 25 devices Internet access; however, we only have one public IP address. What can we use to allow multiple private IP addresses to use a single public IP address?

Answer: NAT

Q3) I want to find another host's hardware (MAC) address on the network. Which protocol lets me get this information?

Answer: ARP

Q4) I want to confirm that I can exchange packets with another host as if I am playing 'ping pong'. Which protocol supports this?

Answer: ICMP